

AI Clinical Workflow Copilot

Dataset Requirements, Columns, Row Targets & Synthetic Data Plan

This document translates the supplied product requirements into a practical dataset plan for a client-demo-ready MVP. The source project requires consultation capture/transcription, medical information extraction, structured documentation, referrals, follow-up/discharge outputs, clinical coding and healthcare analytics. The recommended design uses linked datasets rather than one very large dataset.

1. Recommended Dataset Summary

#	Dataset	Columns	Target rows	Primary use
1	Consultation / Transcription	12	2,000	Audio → transcript + clinical context
2	Clinical NER / Entity Annotation	7	20,000–30,000	Symptoms, diagnosis, medicines, allergies, vitals, history
3	SOAP / Clinical Documentation	8	2,000–3,000	Transcript → SOAP / summary / visit note
4	Workflow Outputs	9	2,000–3,000	Referral, follow-up, discharge, prescription summary
5	SNOMED / ICD Coding	5	10,000–20,000	Diagnosis → code suggestions
6	Medication Reference / Prescription	8	5,000–10,000	Medication normalization and summaries
7	Analytics / Encounter Data	12	5,000–10,000	Symptoms, medication and visit analytics

Important: The row targets are engineering recommendations for a strong prototype, not minimum regulatory or clinical-validation requirements. For an MVP, quality, diversity, correct labels and linkage between records matter more than simply maximizing row count.

2. Dataset 1 — Consultation / Transcription

Column	Type	Description
consultation_id	string	Unique consultation identifier
patient_id	string	Synthetic/de-identified patient identifier
doctor_id	string	Synthetic/de-identified clinician identifier
date	date	Consultation date
specialty	category	GP, cardiology, neurology, orthopedics, etc.
audio_path	string	Path/URI for audio; optional for text-only records
audio_duration_sec	number	Audio duration in seconds
transcript	text	Doctor–patient conversation transcript
symptoms	text/list	Symptoms mentioned in consultation
medical_history	text	Relevant history
allergies	text/list	Known allergies
vital_signs	JSON/text	BP, pulse, temperature, SpO2, etc.

Row target: Target: 2,000 consultations. Include multiple specialties, consultation lengths and realistic conversational variation.

3. Dataset 2 — Clinical NER / Entity Annotation

Column	Type	Description
sentence_id	string	Unique sentence identifier
consultation_id	string	Links sentence to consultation
text	text	Clinical sentence
entity	text	Exact entity span
entity_type	category	SYMPTOM, DIAGNOSIS, MEDICATION, ALLERGY, VITAL, HISTORY, PATIENT_INFO
start	integer	Character start position
end	integer	Character end position

Row target: Target: 20,000–30,000 sentences. Prefer several examples for each entity type and multiple ways of expressing the same concept.

4. Dataset 3 — SOAP / Clinical Documentation

Column	Type	Description
consultation_id	string	Links to consultation
transcript	text	Input conversation/transcript
subjective	text	Patient-reported symptoms/history
objective	text	Observed/measured findings
assessment	text	Clinical assessment/working diagnosis
plan	text	Treatment/referral/follow-up plan
consultation_summary	text	Concise clinical summary
patient_visit_note	text	Structured visit documentation

Row target: Target: 2,000–3,000 consultation-note pairs. This is the main supervised/example dataset for structured documentation.

5. Dataset 4 — Clinical Workflow Outputs

Column	Type	Description
consultation_id	string	Links to consultation
clinical_summary	text	Input summary
diagnosis	text	Diagnosis/assessment
symptoms	text/list	Key symptoms
specialist	category	Target referral specialty; nullable when no referral
referral_reason	text	Reason for specialist referral
referral_letter	text	Generated/approved referral letter
followup_instructions	text	Follow-up instructions
discharge_summary	text	Discharge instructions/summary

Row target: Target: 2,000–3,000 records. Include both referral and non-referral cases; do not make every consultation require a referral.

6. Dataset 5 — SNOMED / ICD Coding

Column	Type	Description
diagnosis	text	Common clinical diagnosis term
diagnosis_description	text	Normalized/fully specified description
synonyms	text/list	Alternative terms/phrasing
snomed_code	string	SNOMED CT concept identifier
icd_code	string	ICD code where applicable

Row target: Target: 10,000–20,000 mappings. This can be built primarily from terminology/reference sources rather than manually authored examples.

7. Dataset 6 — Medication Reference / Prescription

Column	Type	Description
medication_id	string	Unique medication record
consultation_id	string	Links to consultation
medication_name	text	Medication name
generic_name	text	Generic/normalized name
dosage	text	Dose
frequency	text	e.g., once daily
duration	text	Treatment duration
instructions	text	Patient-facing prescription instructions

Row target: Target: 5,000–10,000 records. Include common medicines, dosage variations and different natural-language instructions.

8. Dataset 7 — Analytics / Encounter Data

Column	Type	Description
consultation_id	string	Unique encounter
patient_id	string	Synthetic/de-identified patient ID
date	date	Encounter date
age	integer	Age at encounter
gender	category	Use a project-approved category set
specialty	category	Consultation specialty
symptoms	text/list	Key symptoms
diagnosis	text	Primary diagnosis
medications	text/list	Medications mentioned/prescribed
referral	boolean	Whether referral was generated
followup_required	boolean	Whether follow-up was recommended

Row target: Target: 5,000–10,000 encounters. This dataset can be derived from the consultation/clinical records; it does not have to be independently collected.

9. Which Datasets Can Be Found Openly?

Clinical notes: MIMIC-IV-Note is a major source of de-identified free-text clinical notes. The current PhysioNet page describes 331,794 de-identified discharge summaries and 2,321,355 radiology reports in version 2.2, but access is credentialed and requires training and a data-use agreement. ■cite■turn0search0■turn0search1■

Clinical coding: SNOMED CT has an International Edition with RF2 releases. The U.S. National Library of Medicine currently lists the August 2026 International Edition and provides RF2 download information. Licensing/access conditions should be checked for the intended use. ■cite■turn0search4■

SNOMED entity-linking examples: PhysioNet also lists a 2026 SNOMED CT Entity Linking Challenge containing 272 discharge notes annotated with SNOMED CT concepts. This is useful as an annotation/reference resource but is far smaller than the recommended project target. ■cite■turn0search6■

Audio/transcription: Public clinical conversation/audio datasets are much more constrained than general speech datasets. For a demo, it is safer to use appropriately licensed clinical speech data where available and supplement missing specialty/conversation patterns with synthetic or scripted conversations rather than fabricating claims that a public dataset contains the exact required fields.

10. If a Dataset Is Not Available: Synthetic Data Plan

1. Create a controlled vocabulary first: specialties, symptoms, diagnoses, medications, allergies, vitals, referral departments and common follow-up scenarios.
2. Generate synthetic patient IDs and clinician IDs. Never use real names, phone numbers, addresses or other identifiers.
3. Generate consultation scenarios from templates: complaint → history → symptoms → examination/vitals → assessment → plan.
4. Generate multiple linguistic variations for the same clinical concept so the NLP model does not learn one fixed sentence pattern.
5. For NER, create the sentence plus exact entity spans and entity types. Validate that every start/end index matches the text.
6. For SOAP data, keep the transcript as the input and generate a clinically coherent Subjective, Objective, Assessment and Plan target.
7. For workflow data, create both positive and negative referral cases, different specialist types, follow-up periods and discharge situations.
8. For coding data, use authoritative terminology/reference sources for codes. Do not invent SNOMED identifiers.
9. Run validation checks: missing values, duplicate IDs, invalid code formats, inconsistent diagnosis/medication combinations and impossible vital signs.
10. Keep a `source\_type` field such as `public`, `synthetic`, `manually\_reviewed` and `generated` so the provenance of every record is clear.

11. Synthetic Data Example Structure

specialty	transcript	symptoms	diagnosis	medications
Neurology	Patient reports dizziness for three days...	dizziness; nausea	suspected vestibular disorder	example medication
Orthopedics	Patient reports knee pain after...	knee pain; swelling	knee injury	example medication
General Practice	Patient reports seasonal cough...	cough	upper respiratory symptoms	example medication

The examples above are schema examples only. They should not be presented as real patient cases or used as clinical guidance.

12. Final Recommended Build

Best practical setup: Start with 2,000 linked consultation records. From those records, produce approximately 20,000–30,000 NER-labelled sentences, 2,000–3,000 documentation examples and 2,000–3,000 workflow outputs. Maintain a separate 10,000–20,000-row clinical terminology/code table and a 5,000–10,000-row medication reference/encounter-derived analytics layer. This gives the project enough breadth to demonstrate the full workflow without creating unnecessary independent datasets.

The supplied specification explicitly describes the workflow as audio upload → transcription → NLP extraction → structured documentation → workflow outputs → doctor review → save/export, and the dashboard includes healthcare analytics. ■filecite■turn0file0■L100-L118■ ■filecite■turn0file0■L74-L83■